

JEE MAIN CRASH COURSE

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- ✓ **LATEST NTA PATTERN**
- ✓ **CONCEPT VIDEOS**
- ✓ **LIVE DOUBT SESSIONS**
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Designed by
Math Maestro

Anup Sir



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TARGETING JEE ADVANCED 2020

- ✓ CONCEPT BUILDING VIDEOS
- ✓ ADVANCED LEVEL VIDEOS
- ✓ 2 LEVELS OF SHEETS,
1 WITH ALL TYPE OF QUES.
- ✓ LIVE DISCUSSION OF
PREVIOUS YEAR QUESTIONS

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JEE Mains 2020 Jan Chapter wise Question Bank

Parabola

Q1

If $y = mx + 4$ is common tangent to parabolas $y^2 = 4x$ and $x^2 = 2by$. Then value of b is

यदि $y = mx + 4$ परवलय $y^2 = 4x$ तथा $x^2 = 2by$ की उभयनिष्ठ स्पर्श रेखा है तो b का मान है—

(1) -64

(2) -32

(3) -128

(4) 16

7th Jan Morning

Sol

(3)

$$y = mx + 4 \quad \dots\dots(i)$$

$$y^2 = 4x \text{ tangent की स्पर्श रेखा } y = mx + \frac{a}{m} \Rightarrow y = mx + \frac{1}{m} \quad \dots\dots(ii)$$

from (i) and (ii) से

$$4 = \frac{1}{m} \Rightarrow m = \frac{1}{4}$$

So line $y = \frac{1}{4}x + 4$ is also tangent to parabola $x^2 = 2by$, so solve

अतः रेखा $y = \frac{1}{4}x + 4$ परवलय $x^2 = 2by$ की स्पर्श रेखा है अतः

$$x^2 = 2b \left(\frac{x+16}{4} \right)$$

$$\Rightarrow 2x^2 - bx - 16b = 0 \quad \Rightarrow D = 0 \Rightarrow b^2 - 4 \times 2 \times (-16b) = 0$$

$$\Rightarrow b^2 + 32 \times 4b = 0$$

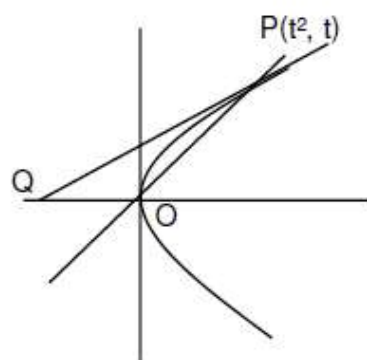
$$b = -128, b = 0 \text{ (not possible असंभव)}$$

02

Let the line $y = mx$ intersects the curve $y^2 = x$ at P and tangent to $y^2 = x$ at P intersects x-axis at Q. If area $(\Delta OPQ) = 4$, find m ($m > 0$)

8th Jan Evening

Sol



$$2ty = x + t^2$$

$$Q(-t^2, 0)$$

$$\frac{1}{2} \begin{vmatrix} 0 & 0 & 1 \\ t^2 & t & 1 \\ -t^2 & 0 & 1 \end{vmatrix} = 4$$

$$|t|^3 = 8$$

$$t = \pm 2 \quad (t > 0)$$

$$m = \frac{1}{2}$$

03

Let one end of focal chord of parabola $y^2 = 8x$ is $\left(\frac{1}{2}, -2\right)$, then equation of tangent at other end of this focal chord is

माना परवलय $y^2 = 8x$ की नाभिय जीवा के एक सिरे के निर्देशांक $\left(\frac{1}{2}, -2\right)$ है, तो दूसरे सिरे पर खींची गई स्पर्श रेखा का समीकरण ज्ञात कीजिये।

(1) $x + 2y + 8 = 0$

(2) $x + 2y = 8$

(3) $x - 2y = 8$

(4) $x - 2y + 8 = 0$

9th Jan Evening

Sol

(4)

Let माना $\left(\frac{1}{2}, -2\right)$ is $(2t^2, 4t) \Rightarrow t = \frac{-1}{2}$

Parameter of other end of focal chord is 2

$\Rightarrow$ point is $(8, 8)$

$\Rightarrow$ equation of tangent is $8y - 4(x+8) = 0$

$$\Rightarrow 2y - x = 8$$

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